Catalyst. *by Mrs. Rana*

GRADE 9 · GENERAL · TERM 1 · REVEAL MATH INTEGRATED I

Math Revision Booklet

Every lesson of the term: each question type with its method and a solved example, the traps examiners love, and a last look for the night before the exam.

UNIT 1 · MODULE 1

Expressions

1-1 Numerical Expressions 2
1-2 Algebraic Expressions 5
1-3 Properties of Real Numbers 8
1-4 Distributive Property 11

UNIT 2 · MODULE 2

Equations in One Variable

2-1 Writing and Interpreting Equations 14
2-2 Solving One-Step Equations 17
2-3 Solving Multi-Step Equations 20
2-4 Solving Equations with the Variable on Each Side 23
2-5 Solving Equations Involving Absolute Value 26
2-6 Solving Proportions 29
2-7 Using Formulas 32

UNIT 3 · MODULE 4

Linear and Nonlinear Functions

4-1 Graphing Linear Functions 34
4-2 Rate of Change and Slope 37
4-3 Slope-Intercept Form 40
4-4 Transformations of Linear Functions 43
4-5 Arithmetic Sequences 46
4-6 Piecewise and Step Functions 49
4-7 Absolute Value Functions 52

1 · MUST KNOW
Learn each type: steps, then the solved example.

2 · DON'T FALL FOR IT
Read the traps before every test.

3 · SCAN FOR PRACTICE
Every lesson has a QR with endless new questions.

Scan me

The full course: lessons, flashcards, quizzes and timed exams with new numbers every time.

LESS EFFORT. FULL MARK.

Order Matters

Numerical Expressions

Book 1-1 · p.3–12 | 6 question types | Expressions

SCAN FOR PRACTICE

MUST KNOW

TYPE 1 Translate a verbal expression

1. Change each word into its symbol: plus +, minus −, times ×, divided by ÷.
حوّل كل كلمة لرمزها.
2. "The sum of / difference of / product of A and B" is one group → put it in brackets.
"مجموع / فرق / حاصل ضرب" = قوسين.
3. "A less than B" means **B − A**: the order flips.
"أقل من" بتقلب الترتيب.

SOLVED EXAMPLE

Translate *the sum of five and nine divided by two.*

the sum of five and nine → **(5 + 9)** — one group

divided by two → **÷ 2**

Answer: $\frac{5+9}{2}$

TYPE 2 Write an expression for a real-world rule

1. Read the rule and number the steps.
رقّم الخطوات.
2. Write step 1 first. If more steps act on it, put it in brackets.
الخطوة الأولى بين قوسين لو هيتعمل عليها حاجة.
3. The first step in words is not always the first thing you write.
أول كلمة مش دايمًا أول حاجة تتكتب.

SOLVED EXAMPLE

Bowling handicap: subtract the average from 200, multiply by 2, divide by 3. Marley's average is 170.

200 − 170 — step 1

2(200 − 170) — step 2: multiply the difference

$\frac{2(200-170)}{3}$ — step 3: divide

Answer: $\frac{2(200-170)}{3} = 20$

TYPE 3 Evaluate powers

1. Find the base and the exponent.
حدد الأساس والأس.
2. Write the base as a factor "exponent" times.
اكتب الأساس كعامل بعدد الأس.
3. Multiply. 2^4 is not $2 \cdot 4$.
2^4 مش 2 × 4.

SOLVED EXAMPLE

Evaluate 4^5.

$4 \cdot 4 \cdot 4 \cdot 4 \cdot 4$ — 4 used 5 times

$16 \cdot 16 \cdot 4$

Answer: 1024

Order Matters

Numerical Expressions

Book 1-1 · p.3–12 | 6 question types | Expressions

SCAN FOR PRACTICE

MUST KNOW · CONTINUED

TYPE 4 Order of operations

1. Grouping symbols first.
الأقواس أولًا.
2. Then all powers.
بعدين الأسس.
3. Then × and ÷ from left to right.
بعدين الضرب والقسمة من الشمال لليمين.
4. Then + and − from left to right.
وآخر حاجة الجمع والطرح من الشمال لليمين.

SOLVED EXAMPLE

Evaluate $20 - 7 + 8^2 - 7 \cdot 11$.

$20 - 7 + 64 - 7 \cdot 11$	power
$20 - 7 + 64 - 77$	multiply
$13 + 64 - 77$	left to right
$77 - 77$	

Answer: 0

TYPE 5 Grouping symbols and fraction bars

1. Work from the innermost brackets outward: ( ) then [ ].
ابدأ من القوس الداخلي.
2. With a fraction bar, find the top and the bottom first.
احسب البسط والمقام الأول.
3. Divide last.
اقسم في الآخر.

SOLVED EXAMPLE

Evaluate $15 - [10 + (3 - 2)^2] + 6$.

$15 - [10 + 1^2] + 6$	innermost
$15 - [11] + 6$	power, add
$4 + 6$	left to right

Answer: 10

TYPE 6 Write and evaluate a real-world expression

1. For each item: number × price (or points).
لكل حاجة: العدد × السعر.
2. Add the parts. A flat fee is added once.
اجمع، والرسوم الثابتة مرة واحدة.
3. Evaluate with the order of operations and write the unit.
احسب واكتب الوحدة.

SOLVED EXAMPLE

Mellie rolls 2 balls into the 30-point hole, 4 into the 20-point hole and 3 into the 50-point hole. Find her score.

$2 \cdot 30 + 4 \cdot 20 + 3 \cdot 50$	write
$60 + 80 + 150$	multiply

Answer: 290 points

DON'T FALL FOR IT

× WRONG
~~$2^4 = 8$~~

✓ RIGHT
$2^4 = 16$

The exponent is how many times you multiply the base, not a factor.
الأس مش بيتضرب في الأساس.

× WRONG
~~$20 - 7 + 64 - 77 = 20 - 71 - 77$~~

✓ RIGHT
$13 + 64 - 77 = 0$

+ and − go left to right. Do not add first.
الجمع والطرح بالترتيب من الشمال لليمين.

× WRONG
~~25 less than 100 → 25 − 100~~

✓ RIGHT
$100 - 25$

"less than" flips the order.
"أقل من" بتقلب الترتيب.

Last Look

Numerical Expressions

THE WHOLE LESSON ON ONE PAGE

1 Words → symbols
plus +, less than (flip the order) −, times ×, divided by ÷. "The sum of…" makes a group.

the sum of 5 and 9 divided by 2 → $\frac{5+9}{2}$

"مجموع" أو "فرق" = مجموعة بين قوسين.

2 Real-world rule
Write the steps in order. Keep earlier steps in brackets.

$\frac{2(200-170)}{3}$

اكتب الخطوات بالترتيب، والخطوة الأولى بين قوسين.

3 Powers
Base = the number multiplied. Exponent = how many times.

$2^4 = 2 \cdot 2 \cdot 2 \cdot 2 = 16$

الأساس يتضرب في نفسه عدد مرات = الأس.

Numerical Expressions

4 Order of operations
Grouping → powers → × ÷ left to right → + − left to right.

$20 - 7 + 8^2 - 7 \cdot 11 = 0$

أقواس، أسس، ضرب وقسمة، جمع وطرح، من الشمال لليمين.

5 Grouping & fraction bars
Innermost bracket first. A fraction bar groups the top and the bottom.

$\frac{(4+5)^2}{3(7-4)} = 9$

خط الكسر: احسب البسط والمقام كل واحد لوحده.

6 Write & evaluate
Number × price for each item, then add. A flat fee is added once.

$50 + 25(2+3) = 175$

اضرب العدد في السعر، والرسوم الثابتة مرة واحدة.

KEY WORDS · الكلمات المهمة

numerical expression	تعبير عددي
power	قوة
base	الأساس
exponent	الأس
evaluate	أوجد القيمة
order of operations	ترتيب العمليات
grouping symbols	رموز التجميع
product · quotient	حاصل الضرب · حاصل القسمة

NIGHT-BEFORE CHECK

Q1 · BOOK STYLE
Which is a numerical expression for *seven times fifteen minus two times nine*?

A $7(15-2) \cdot 9$
B $(7+15) - 2 \cdot 9$
C $7 \cdot (15 - 2 \cdot 9)$
D $7 \cdot 15 - 2 \cdot 9$

Q2 · BOOK STYLE
Evaluate $20 - 7 + 8^2 - 7 \cdot 11$.

Q3 · LIKE BOOK P.4
Which expression changes 37 °C to °F? (Multiply by 9, divide by 5, add 32.)

A $\frac{37 \cdot 5}{9} + 32$
B $\frac{37 \cdot 9}{5 + 32}$
C $\frac{37 \cdot 9}{5} \cdot 32$
D $\frac{37 \cdot 9}{5} + 32$

Q4 · BOOK STYLE
Evaluate $\frac{(1+6)9}{5^2 - 4}$.

Answers: Q1 D · Q2 0 · Q3 D · Q4 3

Let x Talk

Algebraic Expressions

Book 1-2 · p.13–22 | 5 question types | Expressions

SCAN FOR PRACTICE

MUST KNOW

TYPE 1 Write a verbal expression

1. Read the expression from left to right.
اقرأ من الشمال لليمين.
2. x^2 "x squared", x^3 "x cubed", x^n "x to the nth power".
الأسس ليها أسماء.
3. A fraction is "the quotient of... and..." or "divided by".
الكسر = قسمة.

SOLVED EXAMPLE

Write a verbal expression for $a^4 + \frac{6b}{7}$.

a^4 → a to the fourth power — first term

$+$ → plus · $\frac{6b}{7}$ → the quotient of 6 times b and 7 — second term

Answer: a to the fourth power plus the quotient of 6 times b and 7

TYPE 2 Write an algebraic expression

1. Find the key words and their operations.
حدد الكلمات المفتاحية.
2. "times the quantity / the sum of" → brackets.
"times the quantity" = قوسين.
3. "less than" and "subtracted from" flip the order.
بيقلبوا الترتيب.

SOLVED EXAMPLE

Write *2 times the quantity y plus 11.*

times → ×, the quantity → ( ) — key words

2(y + 11)

Answer: 2(y + 11)

TYPE 3 Define a variable and write an expression

1. Find each unknown and give it a letter.
كل مجهول ياخد حرف.
2. Multiply each variable by its value (points, price, rate).
المتغير × قيمته.
3. Add the parts. A fixed fee is added once.
اجمع، والثابت مرة واحدة.

SOLVED EXAMPLE

3 points for a win, 1 for a tie, 0 for a loss. Write the total points.

w = wins, t = ties, z = losses — define

3w + 1t + 0z — points

3w + t — simplify

Answer: 3w + t

Let x Talk

Algebraic Expressions

Book 1-2 · p.13–22 | 5 question types | Expressions

SCAN FOR PRACTICE

MUST KNOW · CONTINUED

TYPE 4 Evaluate an algebraic expression

1. Replace each variable with its value, in brackets.
   عوّض بين قوسين.
2. Use the order of operations.
   ترتيب العمليات.
3. Watch negative values and powers.
   خلي بالك من السالب والأسس.

SOLVED EXAMPLE

Evaluate $a^2(3b - a + 5) \div c$ if a = 2, b = 6, c = 4.

$2^2(3 \cdot 6 - 2 + 5) \div 4$	substitute
$2^2(21) \div 4$	brackets
$4(21) \div 4 = 84 \div 4$	power, multiply

Answer: 21

TYPE 5 Write and evaluate (real world)

1. Write the expression with a variable.
   اكتب التعبير.
2. Substitute the given value.
   عوّض.
3. Evaluate and write the unit.
   احسب واكتب الوحدة.

SOLVED EXAMPLE

The surface area of a sphere is 4 times π times the radius squared. Find it for r = 25.4 cm.

$A = 4\pi r^2$	write
$4\pi(25.4)^2 = 4\pi(645.16)$	substitute
$= 2580.64\pi$	

Answer: ≈ 8107.32 cm²

DON'T FALL FOR IT

× WRONG
~~5 less than x → 5 − x~~

✓ RIGHT
x − 5

"less than" flips the order.
"أقل من" بتقلب الترتيب.

× WRONG
~~$3a^2$ with a = 4 → $(12)^2 = 144$~~

✓ RIGHT
3(16) = 48

The power belongs to **a** only, not to **3a**.
الأس على a بس.

× WRONG
~~2 times the quantity y plus 11 → 2y + 11~~

✓ RIGHT
2(y + 11)

"the quantity" means brackets.
"the quantity" = قوسين.

Last Look

Algebraic Expressions

THE WHOLE LESSON ON ONE PAGE

1 Algebra → words
Read in order. x^2 "squared", x^3 "cubed". A bracket = "the sum / difference / quantity".
$5x^3 + 2$ → five times x cubed plus two
اقرأ بالترتيب، والقوس = "مجموع/فرق".

2 Words → algebra
more than, increased by +; less than, decreased by −; product, times ×; quotient ÷.
18 decreased by 3 times d → 18 − 3d
"أقل من" بتقلب الترتيب.

3 Define a variable
Pick a letter for each unknown, then write the rule.
w wins, t ties → 3w + t
حدد حرف لكل مجهول.

Algebraic Expressions

4 Evaluate (substitute)
Replace each letter with its value in brackets, then use the order of operations.
$a^2(3b - a + 5) \div c = 21$
عوّض بين قوسين وبعدين ترتيب العمليات.

5 Write & evaluate
Fixed amount once + rate × variable. Then substitute.
5 + 3k, k = 8 → 29
الثابت مرة واحدة + المعدل × المتغير.

KEY WORDS · الكلمات المهمة

variable	متغير
algebraic expression	تعبير جبري
term	حد
variable term	حد متغير
constant term	حد ثابت
define a variable	تعريف المتغير
Substitution Property	خاصية التعويض

NIGHT-BEFORE CHECK

Q1 · LIKE BOOK P.13
Which verbal expression represents $\frac{2}{5}m^2$?

A two fifths times m squared
B two fifths times m cubed
C the quotient of two and five times the product of m and two
D two fifths times the product of m and two

Q2 · BOOK STYLE
Which is an algebraic expression for *6 more than twice m*?

A $2m + 6$
B $2(m + 6)$
C $2m - 6$
D $6m + 2$

Q3 · BOOK STYLE
Evaluate $a^2(3b - a + 5) \div c$ if $a = 2$, $b = 6$, $c = 4$.

Q4 · BOOK STYLE
Evaluate $r^2 + (g^3 - 8)^5$ if $g = 2$ and $r = 3$.

Answers: Q1 A · Q2 A · Q3 21 · Q4 9

Rules That Never Break

Properties of Real Numbers

Book 1-3 · p.23–34 | 5 question types | Expressions

SCAN FOR PRACTICE

MUST KNOW

TYPE 1 Identify the property of equality

1. Same thing on both sides → Reflexive.
   نفس الشيء على الطرفين ← انعكاس.
2. "If a = b, then b = a" → Symmetric.
   تبديل الطرفين ← تماثل.
3. Two equal statements linked by a middle quantity → Transitive.
   سلسلة ← تعدي.

SOLVED EXAMPLE

If 13 + 25 = 38, then 38 = 13 + 25. Which property?

The two sides swapped places.

Answer: Symmetric Property

TYPE 2 Complete a statement using a property

1. Read which property you must use.
   اقرأ الخاصية المطلوبة.
2. Reflexive: copy the same quantity. Transitive: jump to the last quantity in the chain.
   الانعكاس: نفس الشيء. التعدي: آخر حلقة في السلسلة.

SOLVED EXAMPLE

If 24 + 11 = 9 + 26 and 9 + 26 = z, then 24 + 11 = ? (Transitive)

24 + 11 = 9 + 26 = z — chain

Answer: z

TYPE 3 Name the property used

1. + 0 → additive identity; + opposite → additive inverse.
   0+ محايد جمعي، + المعكوس = 0.
2. ·1 → multiplicative identity; × reciprocal → multiplicative inverse; × 0 → zero.
   1× محايد ضربي، × المقلوب = 1.
3. Order changed → commutative; brackets moved → associative.
   ترتيب اتغير ← إبدال، أقواس اتحركت ← تجميع.

SOLVED EXAMPLE

Name the property in each step:
4·1 + 0 − 4 + 3 = 4 + 0 − 4 + 3 = 4 − 4 + 3 = 0 + 3 = 3.

4·1 = 4	Multiplicative Identity
4 + 0 = 4	Additive Identity
4 − 4 = 0	Additive Inverse
0 + 3 = 3	Additive Identity

Four steps, four properties

TYPE 4 Evaluate using the properties

1. Look for pairs that make 0 (opposites) or 1 (reciprocals).
   دوّر على أزواج بتدي 0 أو 1.
2. A product with 0 is 0, but only what is multiplied by 0.
   الصفر بيصفّر اللي مضروب فيه بس.
3. Finish with the identities.
   كمّل بالمحايد.

SOLVED EXAMPLE

Evaluate $4 - 2^2 + 8(2)$.

4 − 4 + 16	power, multiply
0 + 16	additive inverses
16	additive identity

Answer: 16

Rules That Never Break

Properties of Real Numbers

Book 1-3 · p.23–34 | 5 question types | Expressions

SCAN FOR PRACTICE

MUST KNOW · CONTINUED

TYPE 5 Commutative and associative (mental math)

1. Find numbers that make an easy total (10s, 100s) or a product of 1.
   دوّر على أعداد مجموعها سهل.
2. Commutative: move them next to each other.
   الإبدال: قرّبهم من بعض.
3. Associative: put brackets around them. Only when all operations are + or all are ×.
   التجميع: قوس حواليهم، بشرط نفس العملية.

SOLVED EXAMPLE

Jade spent 32, 27, 28 and 33 AED. Find the total.

32 + 28 + 27 + 33	commutative
(32 + 28) + (27 + 33)	associative
60 + 60	

Answer: AED 120

DON'T FALL FOR IT

× WRONG
~~If a = b, then b = a is Reflexive~~

✓ RIGHT
Symmetric

Reflexive is **a = a**. Swapping sides is Symmetric.
التبديل = تماثل، مش انعكاس.

× WRONG
~~5 + (9 − 2) · 0 = 0~~

✓ RIGHT
5 + 0 = 5

Zero only cancels what it multiplies.
الصفر بيصفّر اللي مضروب فيه بس.

× WRONG
~~3 + 5 · 2 = (3 + 5) · 2 (associative)~~

✓ RIGHT
3 + 10 = 13

You cannot regroup when + and × are mixed.
التجميع لعملية واحدة بس.

Last Look

Properties of Real Numbers

THE WHOLE LESSON ON ONE PAGE

1 Reflexive
A quantity equals itself.
$a = a$
الشيء يساوي نفسه.

2 Symmetric
The two sides swap.
if $a = b$, then $b = a$
الطرفين يتبدلوا.

3 Transitive
A chain of equals.
$a = b, b = c \rightarrow a = c$
سلسلة: أ = ب وب = ج إذن أ = ج.

Properties of Real Numbers

4 Identities & inverses
$a + 0 = a \cdot a + (-a) = 0 \cdot a \cdot 1 = a \cdot \frac{a}{b} \cdot \frac{b}{a} = 1$
$\frac{3}{2} \cdot \frac{2}{3} \cdot 7 = 7$
المحايد الجمعي 0، والضربي 1.

5 Zero
Anything times 0 is 0, but check the brackets.
$[4 - 3(2) + 7] \cdot 0 = 0$
أي حاجة × صفر = صفر.

6 Commutative & associative
Change the order or move the brackets. Only for + or only for ×.
$(32 + 28) + (27 + 33) = 120$
الإبدال يغير الترتيب، والتجميع يحرك الأقواس.

KEY WORDS · الكلمات المهمة

Reflexive Property	خاصية الانعكاس
Symmetric Property	خاصية التماثل
Transitive Property	خاصية التعدي
additive identity	المحايد الجمعي
additive inverses	معكوسان جمعيان
multiplicative identity	المحايد الضربي
multiplicative inverses (reciprocals)	معكوسان ضربيان
Commutative Property	خاصية الإبدال
Associative Property	خاصية التجميع

NIGHT-BEFORE CHECK

Q1 · BOOK STYLE
Which property is shown? If $13 + 25 = 38$, then $38 = 13 + 25$.
A Reflexive
B Transitive
C Symmetric
D Commutative

Q2 · BOOK STYLE
If $43 + 9 = 10 + 42$ and $10 + 42 = 52$, then $43 + 9 = \ldots$
A $10 + 42$
B 52
C $9 + 43$
D 43

Q3 · BOOK STYLE
Evaluate $13 + 2^2 + 0$.

Q4 · BOOK STYLE
Evaluate $(13 - 1) \cdot 0 + \frac{3}{4} \cdot \frac{4}{3}$.

Answers: Q1 C · Q2 B · Q3 17 · Q4 1

Share It Out

Distributive Property

Book 1-4 · p.35–44 | 6 question types | Expressions

SCAN FOR PRACTICE

MUST KNOW

TYPE 1 Distributive Property with numbers

1. Write the hard number as an easy sum or difference: 99 = 100 − 1.
   اكتب العدد كمجموع أو فرق سهل.
2. Multiply each part.
   اضرب كل جزء.
3. Add or subtract.
   اجمع أو اطرح.

SOLVED EXAMPLE

Find 4 · 1002.

4(1000 + 2) — 1002 = 1000 + 2

4000 + 8 — distribute

Answer: 4008

TYPE 2 Real-world totals

1. Write the total as rate × (sum of parts) or as a sum of products.
   الكلي = المعدل × مجموع الأجزاء.
2. Both forms are equivalent.
   الشكلين متكافئين.

SOLVED EXAMPLE

A team practises 5 days a week: 15 min stretching, 45 min laps, 30 min weights. Minutes per week?

5(15 + 45 + 30) or 5(15) + 5(45) + 5(30) — equivalent

5(90)

Answer: 450 minutes

TYPE 3 Distribute from the left

1. Multiply the outside number by each term inside.
   اضرب في كل حد.
2. A negative outside changes every sign inside.
   السالب يغير كل الإشارات.

SOLVED EXAMPLE

Simplify 4(5x − 7).

4 · 5x − 4 · 7 — distribute

20x − 28 — multiply

Answer: 20x − 28

TYPE 4 Distribute from the right

1. Multiply each term inside by the number on the right.
   اضرب كل حد في الرقم اللي على اليمين.
2. Keep the exponents. $6 \cdot 3y^2 = 18y^2$, not $18y^3$.
   الأسس ثابتة.

SOLVED EXAMPLE

Simplify $(3y^2 + y - 8)6$.

$6(3y^2) + 6(y) + 6(-8)$ — distribute

$18y^2 + 6y - 48$ — multiply

Answer: $18y^2 + 6y - 48$

Share It Out

Distributive Property

Book 1-4 · p.35–44 | 6 question types | Expressions

SCAN FOR PRACTICE

MUST KNOW · CONTINUED

TYPE 5 Combine like terms

1. Like terms: same variable AND same exponent.
   نفس المتغير ونفس الأس.
2. Add the coefficients; the variable part stays.
   اجمع المعاملات بس.
3. If no like terms, it is already simplified.
   لو مفيش، يبقى في أبسط صورة.

SOLVED EXAMPLE

Simplify $4b^2 + 9b - 3b$.

$4b^2 + (9 - 3)b$	distributive
$4b^2 + 6b$	

Answer: $4b^2 + 6b$

TYPE 6 Write and simplify expressions

1. Translate the words into brackets.
   ترجم لأقواس.
2. Distribute every bracket (careful with −).
   وزّع كل قوس.
3. Use the Commutative Property to bring like terms together, then combine.
   رتّب واجمع.

SOLVED EXAMPLE

Three times the sum of 2x and 3y decreased by twice the difference of 4x and y.

$3(2x + 3y) - 2(4x - y)$	write
$6x + 9y - 8x + 2y$	distribute
$6x - 8x + 9y + 2y$	commutative

Answer: $-2x + 11y$

DON'T FALL FOR IT

× WRONG	✓ RIGHT	
~~$-4(3k + 7p) = -12k + 28p$~~	$-12k - 28p$	The negative multiplies every term. السالب على كل الحدود.
~~$4b^2 + 6b = 10b^3$~~	$4b^2 + 6b$	b^2 and b are not like terms. أسس مختلفة = مش متشابهة.
~~$5(15 + 45 + 30) = 5(15) + 45 + 30$~~	$5(15) + 5(45) + 5(30)$	Multiply every term in the bracket. اضرب في كل حد.

Last Look

Distributive Property

THE WHOLE LESSON ON ONE PAGE

1 The rule
a(b + c) = ab + ac and **a(b − c) = ab − ac**. Works from the right too.

4(9 + 3) = 36 + 12 = 48

اضرب في كل حد جوه القوس.

2 Mental math
Write the number as a sum or difference of easy numbers.

5 · 99 = 5(100 − 1) = 495

100 − 1 = 99.

3 Distribute from the left
Multiply the outside number by each term. Signs matter.

4(5x − 7) = 20x − 28

الإشارة بتتضرب برضه.

Distributive Property

4 Distribute from the right
(b + c)a = ba + ca. Exponents stay the same.

$(3y^2 + y - 8)6 = 18y^2 + 6y - 48$

من اليمين نفس الفكرة.

5 Combine like terms
Same variable and same exponent: add the coefficients.

$4b^2 + 9b - 3b = 4b^2 + 6b$

b^2 و b مش متشابهين.

6 Write & simplify
Translate, distribute, regroup (commutative), combine.

3(2x + 3y) − 2(4x − y) = −2x + 11y

ترجم، وزّع، رتّب، اجمع.

KEY WORDS · الكلمات المهمة

Distributive Property	خاصية التوزيع
coefficient	المعامل
like terms	حدود متشابهة
simplest form	أبسط صورة
equivalent expressions	تعبيرات متكافئة

NIGHT-BEFORE CHECK

Q1 · BOOK STYLE
Use the Distributive Property to find **5 · 99**.

Q2 · BOOK STYLE
Simplify **4(5x − 7)**.

A **20x − 28**
B **20x + 28**
C **20x − 7**
D **9x − 11**

Q3 · BOOK STYLE
Simplify **$(3y^2 + y - 8)6$**.

A **$18y^2 + y - 8$**
B **$18y^3 + 6y - 48$**
C **$18y^2 + 6y - 48$**
D **$18y^2 + 6y - 8$**

Q4 · BOOK STYLE
Simplify **3(2x + 3y) − 2(4x − y)**.

A **−2x + 11y**
B **−2x + 7y**
C **14x + 11y**
D **−2x + 9y**

Answers: Q1 495 · Q2 A · Q3 C · Q4 A

Say It in Math

Writing and Interpreting Equations

Book 2-1 · p.65–74 | 5 question types | Equations in One Variable

SCAN FOR PRACTICE

MUST KNOW

TYPE 1 Write an equation for a sentence

1. Find the word that means "=" (is, equals, is the same as).
دوّر على "is".
2. Translate the left side and the right side.
ترجم كل طرف.
3. "the quotient of 7 and x" is $\frac{7}{x}$: first number on top.
أول رقم في البسط.

SOLVED EXAMPLE

Twenty minus the quotient of 7 and x is the same as twice x.

$20 - \frac{7}{x}$ — left side

$= 2x$ — "is the same as twice x"

Answer: $20 - \frac{7}{x} = 2x$

TYPE 2 Write an equation for a situation

1. Let a variable stand for the unknown.
عرّف المتغير.
2. Find how the parts add up to the total.
الأجزاء = الكل.
3. Write the equation.
اكتب المعادلة.

SOLVED EXAMPLE

799 teens surveyed: 430 say "mostly kind", 193 do not, the rest n are not on social media.

n = teens not on a social network — define

(430 + 193) + n = 799 — parts = total

Answer: (430 + 193) + n = 799

TYPE 3 Translate a sentence into a formula

1. Give each quantity a letter.
حرف لكل كمية.
2. "twice the sum" → add first, then × 2.
"ضعف المجموع" = 2(+).
3. "the product of... divided by..." → a fraction.
الضرب ثم القسمة = كسر.

SOLVED EXAMPLE

The perimeter of a rectangle is twice the sum of the length and the width.

P, \ell, w — variables

twice the sum → 2(\ell + w)

Answer: P = 2(\ell + w)

TYPE 4 Write a sentence for an equation

1. Read the equation from left to right.
اقرأ بالترتيب.
2. A bracket becomes "the quantity..." or "the sum / difference of...".
القوس = the quantity.
3. "=" becomes "equals" or "is".
= تبقى equals.

SOLVED EXAMPLE

Write a sentence for 3(y + 1) = 12.

3 is three times the bracket — "three times"

(y + 1) → the quantity y plus one

Three times the quantity y plus one equals twelve.

Say It in Math

Writing and Interpreting Equations

Book 2-1 · p.65–74 | 5 question types | Equations in One Variable

SCAN FOR PRACTICE

MUST KNOW · CONTINUED

TYPE 5 Interpret an equation

1. Name what each variable stands for.
   كل متغير بيمثل إيه.
2. Explain what each term means in the situation.
   كل حد معناه إيه.
3. Think about constraints (e.g. no negative numbers of items).
   القيود: مفيش أعداد سالبة.

SOLVED EXAMPLE

Interpret $S = 2\ell w + 2\ell h + 2wh$ for a rectangular prism.

$2\ell w$	top + bottom
$2\ell h$	front + back
$2wh$	left + right

S = the sum of the areas of all 6 faces

DON'T FALL FOR IT

× WRONG
~~four times a number less 10 → 4(x − 10)~~

✓ RIGHT
$4x - 10$

Only "four times the quantity…" needs brackets.
مفيش قوس = "quantity" مفيش.

× WRONG
~~the quotient of 7 and x → $\frac{x}{7}$~~

✓ RIGHT
$\frac{7}{x}$

The first number is on top.
الأول في البسط.

× WRONG
~~3(y + 1) = 12 → three times y plus one is twelve~~

✓ RIGHT
three times the quantity y plus one

Without "the quantity" it means **3y + 1**.
لازم "the quantity".

Last Look

Writing and Interpreting Equations

THE WHOLE LESSON ON ONE PAGE

1 Sentence → equation
"is, equals, is the same as" → =. Translate each side.

Twenty minus the quotient of 7 and x is the same as twice x → $20 - \frac{7}{x} = 2x$

"is" = يساوي.

2 Real-world equation
1 define a variable · 2 list the givens · 3 write the sentence as an equation.

(430 + 193) + n = 799

عرّف، حدد المعطيات، اكتب المعادلة.

Equations

3 Formula
Several variables. "twice the sum" → 2(+).

P = 2(\ell + w)

الصيغة فيها أكثر من متغير.

4 Equation → sentence
Brackets: say "the quantity / the sum of" right after the multiplier.

3(y + 1) = 12 → three times the quantity y plus one equals twelve

القوس بعد الرقم مباشرة.

5 Interpret
Explain what each part means in the situation. Constraints limit the solutions.

2\ell h = front + back faces

فسّر كل جزء في المسألة.

KEY WORDS · الكلمات المهمة

equation	معادلة
constraint	قيد
formula	صيغة / قانون

NIGHT-BEFORE CHECK

Q1 · LIKE BOOK P.65
Which equation represents *four times a number less 10 is equal to 16*?

A $4x - 10 = 16$
B $4(x - 10) = 16$
C $4x + 10 = 16$
D $10 - 4x = 16$

Q2 · LIKE BOOK P.66
Etu has read 12 of the 32 chapters. He plans to read **c** chapters each day for 8 days. Which equation fits?

A $12 + 8c = 32$
B $12 + \frac{c}{8} = 32$
C $12 - \frac{c}{8} = 32$
D $12 - 8c = 32$

Q3 · LIKE BOOK P.67
Which formula gives the horsepower: the product of the speed **M** and the torque **T** divided by 5252?

A $H = \frac{5252M}{T}$
B $H = \frac{5252}{MT}$
C $H = \frac{MT}{5252}$
D $H = \frac{M}{5252T}$

Q4 · LIKE BOOK P.68
Which sentence represents $(y + 3)^2 = 25$?

A y squared plus 3 is 25.
B y plus 3 is 25.
C The quantity y plus 3 squared is 25.
D y plus 3 squared is 25.

Answers: Q1 A · Q2 A · Q3 C · Q4 C

One Move Wins

Solving One-Step Equations

Book 2-2 · p.75–84 | 3 question types | Equations in One Variable

SCAN FOR PRACTICE

MUST KNOW

TYPE 1 Solve by adding or subtracting

1. Find the number next to the variable.
   الرقم اللي مع المتغير.
2. Do the opposite operation to both sides.
   العملية العكسية للطرفين.
3. Check: substitute the answer.
   عوّض للتأكد.

SOLVED EXAMPLE

Solve $g - 25 = 113$.

$g - 25 + 25 = 113 + 25$ — add 25

$g = 138$

Check: $138 - 25 = 113$ ✓

Answer: $g = 138$

TYPE 2 Solve by multiplying or dividing

1. Variable times a number → divide both sides by it.
   اقسم على المعامل.
2. Variable divided by a number → multiply both sides by it.
   اضرب في المقام.
3. Fraction in front → multiply by its reciprocal.
   الكسر → المقلوب.

SOLVED EXAMPLE

Solve $\frac{3}{8}x = \frac{9}{4}$.

$\frac{8}{3} \cdot \frac{3}{8}x = \frac{8}{3} \cdot \frac{9}{4}$ — reciprocal

$x = \frac{72}{12}$

Answer: $x = 6$

TYPE 3 Write and solve a one-step equation

1. Define the variable.
   عرّف المتغير.
2. Write the equation from the words.
   اكتب المعادلة.
3. Solve and check that the answer makes sense.
   حل وراجع المنطق.

SOLVED EXAMPLE

96 sophomores, or $\frac{2}{3}$ of the class, go to the game. How many are in the class?

$\frac{2}{3}n = 96$ — write

$n = 96 \cdot \frac{3}{2}$ — reciprocal

Answer: 144 sophomores (the whole is bigger than the part ✓)

One Move Wins

Solving One-Step Equations

Book 2-2 · p.75–84 | 3 question types | Equations in One Variable

SCAN FOR PRACTICE

DON'T FALL FOR IT

× WRONG

$42 = -14y \rightarrow y = 3$

✓ RIGHT

$y = -3$

Divide by −14, keeping the negative sign.

اقسم بالإشارة.

× WRONG

$\frac{3}{8}x = \frac{9}{4} \rightarrow x = \frac{9}{4} \cdot \frac{3}{8}$

✓ RIGHT

$x = \frac{9}{4} \cdot \frac{8}{3} = 6$

Multiply by the reciprocal, not the same fraction.

اضرب في المقلوب.

× WRONG

$g - 25 = 113 \rightarrow g = 88$

✓ RIGHT

$g = 138$

Undo − 25 by adding 25.

الطرح يتعكس بالجمع.

Last Look

Solving One-Step Equations

THE WHOLE LESSON ON ONE PAGE

1 Undo subtraction
Add the same number to each side.
$g - 25 = 113 \rightarrow g = 138$
الطرح نعكسه بالجمع.

2 Undo addition
Subtract the same number from each side.
$27 + k = 30 \rightarrow k = 3$
الجمع نعكسه بالطرح.

3 Undo multiplication
Divide each side by the coefficient (keep its sign).
$42 = -14y \rightarrow y = -3$
القسمة على المعامل بإشارته.

One-Step Equations

4 Fraction coefficient
Multiply each side by the reciprocal.
$\frac{3}{8}x = \frac{9}{4} \rightarrow x = 6$
اضرب في المقلوب.

5 Write & solve
Define the variable, write the equation, solve, check.
$\frac{2}{3}n = 96 \rightarrow n = 144$
عرّف، اكتب، حل، راجع.

KEY WORDS · الكلمات المهمة

solve an equation	حل المعادلة
solution	حل
equivalent equations	معادلات متكافئة
Addition / Subtraction Property of Equality	خاصية الجمع / الطرح للمساواة
Multiplication / Division Property of Equality	خاصية الضرب / القسمة للمساواة

NIGHT-BEFORE CHECK

Q1 · LIKE BOOK P.76
Solve $\frac{2}{3} + w = 1\frac{1}{2}$ and name the property used.

A $\frac{5}{6}$; Subtraction Property of Equality
B $\frac{5}{6}$; Addition Property of Equality
C $\frac{13}{6}$; Subtraction Property of Equality
D $\frac{13}{6}$; Addition Property of Equality

Q2 · BOOK STYLE
Solve $a + 26 = 35$.

Q3 · LIKE BOOK P.77
A male bulldog weighs 15 lb less than a golden retriever. The bulldog weighs 50 lb. Which equation and answer are correct?

A $50 = w - 15$; 35 lb
B $50 = w + 15$; 35 lb
C $50 = 15 - w$; 65 lb
D $50 = w - 15$; 65 lb

Q4 · BOOK STYLE
Solve $42 = -14y$.

Answers: Q1 A · Q2 9 · Q3 D · Q4 −3

Undo, Step by Step

Solving Multi-Step Equations

Book 2-3 · p.85–90 | 4 question types | Equations in One Variable

SCAN FOR PRACTICE

MUST KNOW

TYPE 1 Solve multi-step equations

1. Undo addition or subtraction.
   اعكس الجمع أو الطرح.
2. Undo multiplication or division.
   اعكس الضرب أو القسمة.
3. Check in the original equation.
   عوّض للتأكد.

SOLVED EXAMPLE

Solve $2a - 6 = 4$.

$2a = 10$ — add 6

$a = 5$ — divide by 2

Check: $2(5) - 6 = 4$ ✓

Answer: $a = 5$

TYPE 2 Equations with a fraction bar

1. Multiply each side by the denominator.
   اضرب في المقام.
2. Then undo + or −.
   بعدين اعكس الجمع/الطرح.

SOLVED EXAMPLE

Solve $\frac{n+1}{-2} = 15$.

$n + 1 = -30$ — × (−2)

$n = -31$ — subtract 1

Answer: $n = -31$

TYPE 3 Write and solve a multi-step equation

1. Write the words as an equation (a table helps).
   اكتب المعادلة.
2. Undo the addition, then multiply by the reciprocal.
   اعكس الجمع ثم اضرب في المقلوب.
3. Answer in context.
   اكتب الإجابة بالوحدة.

SOLVED EXAMPLE

The council raised $\frac{2}{5}$ of the trip cost, then AED 150 more. They have AED 630. Find the cost.

$\frac{2}{5}c + 150 = 630$ — write

$\frac{2}{5}c = 480$ — subtract 150

$c = 480 \cdot \frac{5}{2}$ — reciprocal

Answer: AED 1200

Undo, Step by Step

Solving Multi-Step Equations

Book 2-3 · p.85–90 | 4 question types | Equations in One Variable

SCAN FOR PRACTICE

MUST KNOW · CONTINUED

TYPE 4 Equations with letter coefficients

1. Isolate the term with **x** (undo + or −).
   اعزل الحد اللي فيه x.
2. Divide by the letter coefficient (it is not 0).
   اقسم على الحرف.

SOLVED EXAMPLE

Solve $ax + 7 = 5$ for x.

$ax = -2$ — subtract 7

$x = \frac{-2}{a}$ — divide by a

Answer: $x = -\frac{2}{a}$

DON'T FALL FOR IT

× WRONG

~~$2a - 6 = 4 \rightarrow a = 2 - 6$...~~

✓ RIGHT

add 6 first: $2a = 10$

Undo + and − before × and ÷.
الجمع والطرح يتعكسوا الأول.

× WRONG

~~$\frac{n+1}{-2} = 15 \rightarrow n + 1 = 30$~~

✓ RIGHT

$n + 1 = -30$

Multiply by −2, with its sign.
اضرب في المقام بإشارته.

× WRONG

~~$2 - ax = -8 \rightarrow x = -\frac{10}{a}$~~

✓ RIGHT

$x = \frac{10}{a}$

$-ax = -10$; dividing by $-a$ makes it positive.
سالب ÷ سالب = موجب.

Last Look

Solving Multi-Step Equations

THE WHOLE LESSON ON ONE PAGE

Multi-Step Equations

1 Work backward
Undo + or − first, then × or ÷ (the reverse of the order of operations).

$2a - 6 = 4 \rightarrow 2a = 10 \rightarrow a = 5$

اعكس ترتيب العمليات.

2 Fraction bar
Multiply each side by the denominator first.

$\frac{n+1}{-2} = 15 \rightarrow n = -31$

اضرب في المقام الأول.

3 Write & solve
Fraction of the whole + extra = total.

$\frac{2}{5}c + 150 = 630 \rightarrow c = 1200$

الكسر × الكل + الزيادة = المجموع.

4 Letter coefficients
Treat the letter like a number; divide by it (it is not 0).

$ax + 7 = 5 \rightarrow x = \frac{-2}{a}$

الحرف = رقم.

KEY WORDS · الكلمات المهمة

multi-step equation	معادلة متعددة الخطوات
opposite operation	العملية العكسية
reciprocal	المقلوب

NIGHT-BEFORE CHECK

Q1 · BOOK STYLE
Solve $3m + 4 = -11$.

Q2 · BOOK STYLE
Solve $8 = \frac{x-5}{7}$.

Q3 · LIKE BOOK P.87
A store sold $\frac{2}{3}$ of its basketballs, but 8 were returned. Now it has 38. Which equation and answer are correct?

A $\frac{1}{3}b - 8 = 38$; 138
B $\frac{1}{3}b + 8 = 38$; 90
C $\frac{2}{3}b + 8 = 38$; 45
D $\frac{2}{3}b - 8 = 38$; 69

Q4 · LIKE BOOK P.88
Solve $2 - ax = -8$ for x. Assume $a \neq 0$.

A $x = -\frac{10}{a}$
B $x = -\frac{6}{a}$
C $x = \frac{10}{a}$
D $x = \frac{6}{a}$

Answers: Q1 −5 · Q2 61 · Q3 B · Q4 C

Both Sides Count

Solving Equations with the Variable on Each Side

Book 2-4 · p.91–100 | 4 question types | Equations in One Variable

SCAN FOR PRACTICE

MUST KNOW

TYPE 1 Variable on each side

1. Subtract one variable term from both sides.
   اطرح حد المتغير من الطرفين.
2. Move the numbers to the other side.
   انقل الأرقام.
3. Divide by the coefficient and check.
   اقسم وراجع.

SOLVED EXAMPLE

Solve $5 + 7a = 4a - 13$.

$5 + 3a = -13$	subtract $4a$
$3a = -18$	subtract 5
$a = -6$	divide by 3

Answer: $a = -6$

TYPE 2 Equations with grouping symbols

1. Distribute every bracket. A negative outside changes every sign.
   وزّع، والسالب بيقلب الإشارات.
2. Collect variable terms and constants.
   جمّع.
3. Solve.
   حل.

SOLVED EXAMPLE

Solve $7(n - 1) = -2(3 + n)$.

$7n - 7 = -6 - 2n$	distribute
$9n - 7 = -6$	add $2n$
$9n = 1$	add 7

Answer: $n = \frac{1}{9}$

TYPE 3 Write an equation with the variable on each side

1. Write an expression for each person or shape.
   تعبير لكل طرف.
2. Set them equal and solve.
   ساويهم وحل.

SOLVED EXAMPLE

Find x so that a rectangle 5 by $(x + 4)$ and a triangle with base 12 and height $(2x - 6)$ have the same area.

$5(x + 4) = \frac{1}{2}(12)(2x - 6)$	areas equal
$5x + 20 = 12x - 36$	simplify
$56 = 7x$	collect

Answer: $x = 8$

Both Sides Count

Solving Equations with the Variable on Each Side

Book 2-4 · p.91–100 | 4 question types | Equations in One Variable

SCAN FOR PRACTICE

MUST KNOW · CONTINUED

TYPE 4 One solution, no solution or identity

1. Simplify both sides completely.
   بسّط الطرفين.
2. If the variable cancels: a true statement (0 = 0) → identity; a false one (−30 = 20) → no solution.
   لو المتغير اختفى: صح = متطابقة، غلط = لا حل.
3. Otherwise solve for one value.
   غير كده: حل واحد.

SOLVED EXAMPLE

Solve 6(y − 5) = 2(10 + 3y).

6y − 30 = 20 + 6y	distribute
−30 = 20	subtract 6y: false

Answer: no solution (ø)

DON'T FALL FOR IT

× WRONG
~~7n − 7 = −2(3 + n) → 7n − 7 = −6 + n~~

✓ RIGHT
−6 − 2n

Distribute −2 to both terms.
وزّع السالب على الاتنين.

× WRONG
~~−30 = 20 → y = 0~~

✓ RIGHT
no solution

The variable is gone and the statement is false.
جملة غلط = مفيش حل.

× WRONG
~~0 = 0 → x = 0~~

✓ RIGHT
identity (every real number)

A true statement means every value works.
جملة صح = كل الأعداد.

Last Look

Solving Equations with the Variable on Each Side

THE WHOLE LESSON ON ONE PAGE

1 Collect the variable
Subtract the smaller variable term from both sides, then solve.
$5 + 7a = 4a - 13 \rightarrow a = -6$
المتغيرات في طرف.

2 Grouping symbols
Distribute first (keep the negative), then collect.
$7(n - 1) = -2(3 + n) \rightarrow n = \frac{1}{9}$
وزّع بالإشارة.

3 Fraction bar
Multiply both sides by the denominator.
$5y = \frac{12y + 16}{4} \rightarrow y = 2$
اضرب في المقام.

Variable on Each Side

4 Real world
Each side = one person / one shape. Set them equal.
$6.2m = 3.8m + 20$
كل طرف = حاجة.

5 One, none or all
$-30 = 20$ (false) → no solution. $0 = 0$ (always true) → identity.
$6(y - 5) = 2(10 + 3y) \rightarrow$ **no solution**
جملة خطأ = لا حل، جملة صح دايمًا = كل الأعداد.

KEY WORDS · الكلمات المهمة

identity	متطابقة
no solution	لا يوجد حل
one solution	حل واحد

NIGHT-BEFORE CHECK

Q1 · BOOK STYLE
Solve $5 + 7a = 4a - 13$.

Q2 · LIKE BOOK P.94
Solve $7(n - 2) + 8 = 3(n - 4) - 2$.

Q3 · BOOK STYLE
Solve $5y = \frac{12y + 16}{4}$.

Q4 · LIKE BOOK P.96
Which equation is an identity?

A $8(g + 6) = 5g + 3(g + 16)$
B $3w + 2 = 7w$
C $6(y - 5) = 2(10 + 3y)$
D $5x + 5 = 3(5x - 4) - 10x$

Answers: Q1 −6 · Q2 −2 · Q3 2 · Q4 A

Two Ways Home

Solving Equations Involving Absolute Value

Book 2-5 · p.101–108 | 4 question types | Equations in One Variable

SCAN FOR PRACTICE

MUST KNOW

TYPE 1 Solve an absolute value equation

1. Case 1: inside = the number.
   الحالة الأولى: = الرقم.
2. Case 2: inside = the opposite of the number.
   الحالة الثانية: = سالب الرقم.
3. Write both in a set and check each.
   اكتب الاتنين وراجع.

SOLVED EXAMPLE

Solve $|y + 2| = 4$.

$y + 2 = 4 \rightarrow y = 2$ — case 1

$y + 2 = -4 \rightarrow y = -6$ — case 2

Answer: $\{-6, 2\}$

TYPE 2 Isolate the absolute value first

1. Undo + / − and × / ÷ outside the bars.
   اعكس العمليات برّه.
2. Then split into the two cases.
   وبعدين حالتين.

SOLVED EXAMPLE

Solve $3|x + 1| - 2 = 10$.

$3|x + 1| = 12$ — add 2

$|x + 1| = 4$ — divide by 3

$x = 3$ or $x = -5$ — two cases

Answer: $\{-5, 3\}$

TYPE 3 When the answer is ∅

1. Isolate the absolute value.
   اعزل.
2. If it equals a negative number → ∅.
   لو = سالب ← ∅.

SOLVED EXAMPLE

Solve $|3x - 4| = -1$.

A distance cannot be −1.

Answer: ∅

TYPE 4 Write an absolute value equation

1. Find the centre: the average, or the midpoint of the two points.
   المركز.
2. Find the distance: the error, or half the gap.
   المسافة.
3. Write **$|x - \text{centre}| = \text{distance}$**.
   اكتب المعادلة.

SOLVED EXAMPLE

Write an equation for a graph with points at 17 and 27.

midpoint $\frac{17 + 27}{2} = 22$ — centre

$27 - 22 = 5$ — distance

Answer: $|x - 22| = 5$

Two Ways Home

Solving Equations Involving Absolute Value

Book 2-5 · p.101–108 | 4 question types | Equations in One Variable

SCAN FOR PRACTICE

DON'T FALL FOR IT

× WRONG
~~$|y + 2| = 4 \rightarrow y = 2$ only~~

✓ RIGHT
$\{-6, 2\}$

Always check the negative case too.
متنساش الحالة السالبة.

× WRONG
~~$|3x - 4| = -1 \rightarrow x = 1$~~

✓ RIGHT
$\varnothing$

An absolute value is never negative.
المطلق مش ممكن يبقى سالب.

× WRONG
~~$2|x| + 1 = 9 \rightarrow 2x + 1 = \pm 9$~~

✓ RIGHT
$|x| = 4$ first

Isolate the bars before the two cases.
اعزل الأول.

Last Look

Solving Equations Involving Absolute Value

THE WHOLE LESSON ON ONE PAGE

Absolute Value Equations

1 Two cases
$|A| = n$ with $n > 0$ → $A = n$ or $A = -n$.
$|y + 2| = 4$ → $\{-6, 2\}$
حالتين: موجب وسالب.

2 Isolate first
Get $|...|$ alone before splitting into cases.
$2|x - 1| + 3 = 11$ → $|x - 1| = 4$
اعزل المطلق الأول.

3 Negative → ø
Distance cannot be negative.
$|3x - 4| = -1$ → ø
سالب = لا حل.

4 Write from a graph
Centre = midpoint, distance = half the gap.
17 and 27 → $|x - 22| = 5$
المنتصف والمسافة.

5 Give or take
average ± error → $|x - \text{average}| = \text{error}$
$|x - 2000| = 250$ → 1750, 2250
المتوسط ± الخطأ.

KEY WORDS · الكلمات المهمة

absolute value	القيمة المطلقة
empty set ø	المجموعة الخالية
solution set	مجموعة الحل
midpoint	نقطة المنتصف

NIGHT-BEFORE CHECK

Q1 · BOOK STYLE
Solve $|y + 2| = 4$.
A {2}
B {−6, −2}
C {−6, 2}
D {−2, 6}

Q2 · BOOK STYLE
Solve $|3x - 4| = -1$.
A $\{1, \frac{5}{3}\}$
B ø
C all real numbers
D {1}

Q3 · LIKE BOOK P.102
For $|ax + b| = c$, which statement must be true for the solution to be ø?
A **c** is negative
B **b** is negative
C **c** is positive
D **a** is negative

Q4 · LIKE BOOK P.104
Which equation matches a graph with points at −10 and 2?
A $|x - 4| = 6$
B $|x - 1| = 6$
C $|x + 6| = 4$
D $|x + 4| = 6$

Answers: Q1 C · Q2 B · Q3 A · Q4 D

Keep the Ratio

Solving Proportions

Book 2-6 · p.109–116 | 4 question types | Equations in One Variable

SCAN FOR PRACTICE

MUST KNOW

TYPE 1 Solve a proportion

1. Multiply each side by the denominator under the variable, or use cross products.
   اضرب في المقام أو تبادلي.
2. Divide to finish. Round if asked.
   اقسم، وقرّب لو مطلوب.
3. Check: both fractions simplify to the same value.
   راجع.

SOLVED EXAMPLE

Solve $\frac{x}{45} = \frac{15}{25}$.

$x = \frac{45(15)}{25}$ × 45

$x = \frac{675}{25}$

Answer: $x = 27$ (check $\frac{27}{45} = \frac{3}{5}$ ✓)

TYPE 2 Two missing quantities

1. Cross-multiply. Put brackets around a binomial.
   تبادلي، والقوس حوالين الحدين.
2. Distribute, then collect the variable on one side.
   وزّع وجمّع.
3. Solve.
   حل.

SOLVED EXAMPLE

Solve $\frac{x}{9} = \frac{2x - 3}{24}$.

$24x = 9(2x - 3)$ — cross products

$24x = 18x - 27$ — distribute

$6x = -27$ — subtract 18x

Answer: $x = -4.5$

TYPE 3 Proportion with a constant rate

1. Estimate first.
   قدّر الأول.
2. Write $\frac{\text{amount}}{\text{time}} = \frac{\text{amount}}{\text{time}}$ with the same units in the same places.
   نفس الوحدات.
3. Solve and check against the estimate.
   حل وقارن.

SOLVED EXAMPLE

A city sinks 140 cm every 5 years. How much in 12 years?

Estimate: 10 years → 280 cm — estimate

$\frac{140}{5} = \frac{c}{12}$ — write

$c = \frac{12(140)}{5} = \frac{1680}{5}$ — solve

Answer: 336 cm

Keep the Ratio

Solving Proportions

Book 2-6 · p.109–116 | 4 question types | Equations in One Variable

SCAN FOR PRACTICE

MUST KNOW · CONTINUED

TYPE 4 Percent problems with proportions

1. Decide: part to whole ($\frac{\%}{100}$) or part to other part ($\frac{\%}{100-\%}$).

جزء/كل أو جزء/جزء.

2. Write the proportion and solve.

اكتب وحل.

SOLVED EXAMPLE

How many kg of raisins with 14 kg of nuts make a mix that is 30% raisins?

raisins : nuts = 30 : 70 — part : other part

$$\frac{r}{14} = \frac{30}{70}$$

$$r = \frac{14(30)}{70}$$

Answer: 6 kg

DON'T FALL FOR IT

× WRONG

$\frac{x}{9} = \frac{2x-3}{24} \rightarrow 24x = 18x - 3$

✓ RIGHT

$24x = 18x - 27$

Multiply 9 by both terms of **2x − 3**.

اضرب في الحدين.

× WRONG

30% raisins → $\frac{r}{14} = \frac{30}{100}$ (14 = nuts)

✓ RIGHT

$\frac{r}{14} = \frac{30}{70}$

14 kg is the nuts, not the whole mix.

14 دي المكسرات مش الكل.

× WRONG

$\frac{140}{5} = \frac{12}{c}$

✓ RIGHT

$\frac{140}{5} = \frac{c}{12}$

Keep cm on top and years at the bottom on both sides.

نفس الوحدات في نفس المكان.

Last Look

Solving Proportions

THE WHOLE LESSON ON ONE PAGE

Proportions

1 Solve a proportion

Multiply each side by the denominator of x (or cross-multiply).

$\frac{x}{45} = \frac{15}{25} \rightarrow x = 27$

اضرب في المقام أو ضرب تبادلي.

2 Two missing quantities

Cross-multiply, distribute, collect, solve.

$\frac{x}{9} = \frac{2x-3}{24} \rightarrow x = -4.5$

تبادلي، وزّع، جمّع.

3 Constant rate

Same units in the same places: $\frac{\text{amount}}{\text{time}} = \frac{\text{amount}}{\text{time}}$.

$\frac{140}{5} = \frac{c}{12} \rightarrow 336$

نفس الوحدات في نفس الأماكن.

4 Percent mixture

part : whole = % : 100, or part : other part = % : (100 − %).

$\frac{r}{14} = \frac{30}{70} \rightarrow 6$

الجزء/الكل = النسبة/100.

KEY WORDS · الكلمات المهمة

ratio — نسبة	proportion — تناسب
cross products — الضرب التبادلي	rate — معدل

NIGHT-BEFORE CHECK

Q1 · LIKE BOOK P.110

Solve $\frac{n-4}{8} = \frac{3}{2}$.

A 16 B 8
C 9.33 D 4.75

Q2 · LIKE BOOK P.110

Solve $\frac{x}{12} = \frac{2x-5}{18}$.

A 10 B 0.83
C −10 D −3.75

Q3 · BOOK STYLE

Solve $\frac{x}{45} = \frac{15}{25}$.

Q4 · LIKE BOOK P.111

Oscar mixes 8 parts cranberry juice to 3 parts pineapple juice. How many cups of pineapple juice with 48 cups of cranberry?

Answers: Q1 A · Q2 A · Q3 27 · Q4 18

Rearrange the Rule

Using Formulas

Book 2-7 · p.117–128 | 3 question types | Equations in One Variable

SCAN FOR PRACTICE

MUST KNOW

TYPE 1 Solve for a specific variable

1. Circle the variable you want.
   حدد المتغير.
2. Move every other term away from it.
   انقل الباقي.
3. If it appears twice, collect and factor it out; divide by the bracket.
   لو مرتين: عامل مشترك.

SOLVED EXAMPLE

Solve $4p - 7r = pq + 16$ for p.

$4p - pq = 16 + 7r$ — collect p

$p(4 - q) = 16 + 7r$ — factor

$p = \frac{16 + 7r}{4 - q}$ — divide

Answer: $p = \frac{16 + 7r}{4 - q}$

TYPE 2 Use a literal equation

1. Solve the formula for the variable you need.
   حل للمطلوب.
2. Substitute the known values.
   عوّض.
3. Round sensibly (e.g. whole pizzas).
   قرّب بمنطق.

SOLVED EXAMPLE

Trapezoid: area 70 ft², bases 22 ft and 18 ft. Find the height.

$h = \frac{2A}{b_1 + b_2}$ — solve for h

$h = \frac{140}{40}$ — substitute

Answer: 3.5 ft

TYPE 3 Dimensional analysis

1. Write the given amount with its unit.
   اكتب الكمية بوحدتها.
2. Multiply by factors with the old unit at the bottom.
   الوحدة القديمة في المقام.
3. Cancel units and calculate.
   الغي الوحدات واحسب.

SOLVED EXAMPLE

97 285 litres → gallons (1 gal = 3.785 L).

$97285\ \text{L} \cdot \frac{1\ \text{gal}}{3.785\ \text{L}}$ — litres cancel

≈ 25703

Answer: ≈ 25 703 gallons

DON'T FALL FOR IT

× WRONG

~~$2A = h(b_1 + b_2) \rightarrow h = \frac{2A}{b_1} + b_2$~~

✓ RIGHT

$h = \frac{2A}{b_1 + b_2}$

Divide by the whole bracket.
اقسم على القوس كله.

× WRONG

~~$97285\ \text{L} \cdot \frac{3.785\ \text{L}}{1\ \text{gal}}$~~

✓ RIGHT

$\cdot \frac{1\ \text{gal}}{3.785\ \text{L}}$

The unit to cancel must be in the denominator.
الوحدة القديمة في المقام.

× WRONG

~~c ≈ 2.86 → 3 pizzas~~

✓ RIGHT

2 pizzas

3 pizzas would cost more than the budget.
قرّب لتحت عشان الميزانية.

Last Look

Using Formulas

THE WHOLE LESSON ON ONE PAGE

1 Solve for a variable
Treat the other letters like numbers; undo operations around the target.

$5a - 2b = 15 \rightarrow a = 3 + \frac{2b}{5}$

باقي الحروف كأنها أرقام.

2 Variable on each side
Collect the target terms, factor it out, divide by the bracket.

$p = \frac{16 + 7r}{4 - q}$

عامل مشترك.

Formulas

3 Use a formula
Solve for the variable first, then substitute.

$h = \frac{2(70)}{22 + 18} = 3.5$

حل للمتغير وبعدين عوّض.

4 Dimensional analysis
Multiply by conversion factors so the units cancel.

$97285\ \text{L} \cdot \frac{1\ \text{gal}}{3.785\ \text{L}} \approx 25703\ \text{gal}$

الوحدات بتلغي بعض.

KEY WORDS · الكلمات المهمة

formula	صيغة
literal equation	معادلة حرفية
dimensional analysis	تحليل الوحدات
conversion factor	معامل التحويل

NIGHT-BEFORE CHECK

Q1 · BOOK STYLE
Solve $4p - 7r = pq + 16$ for p.

A $p = \frac{16 + 7r}{4 - q}$
B $p = \frac{16 - 7r}{4 - q}$
C $p = \frac{16 + 7r}{4 + q}$
D $p = 16 + 7r - 4q$

Q2 · LIKE BOOK P.119
A jar is a cylinder: $V = \pi r^2 h$. Which formula gives the height?

A $h = \sqrt{\frac{V}{\pi r}}$
B $h = \frac{V}{\pi r^2}$
C $h = \frac{V}{r^2}$
D $h = \pi \cdot r^2$

Q3 · LIKE BOOK P.121
$P = 40c - 300$. Which equation gives the copies c needed for a profit P?

A $c = \frac{P + 300}{40}$
B $c = \frac{P}{40} + 300$
C $c = \frac{P}{40} - 300$
D $c = \frac{P + 40}{300}$

Q4 · BOOK STYLE
A trapezoid has area 70 ft² and bases 22 ft and 18 ft. Find the height in feet.

Answers: Q1 A · Q2 B · Q3 A · Q4 3.5

Draw the Line

Graphing Linear Functions

Book 4-1 · p.209–218 | 4 question types | Linear and Nonlinear Functions

SCAN FOR PRACTICE

MUST KNOW

TYPE 1 Graph by making a table

1. Choose x-values from the domain. اختار قيم x.
2. Substitute to find each y and write ordered pairs. عوّض.
3. Plot and connect with a straight line. ارسم الخط.

SOLVED EXAMPLE

Graph −2x − 3 = y.

x = −4, −2, 0, 1, 3 — choose

(−4, 5), (−2, 1), (0, −3), (1, −5), (3, −9) — substitute

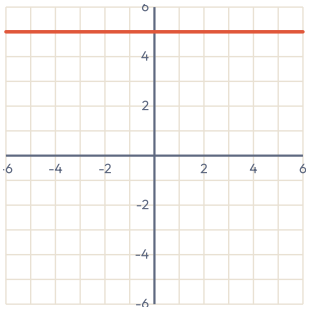
Plot the points and draw the line

TYPE 2 Horizontal and vertical lines

1. **y = a**: every point has the same y → horizontal line. y = رقم ← أفقي.
2. **x = a**: every point has the same x → vertical line, not a function. x = رقم ← رأسي.

SOLVED EXAMPLE

Graph y = 5 and x = −2.



y = 5: (−2, 5), (0, 5), (2, 5) — horizontal

x = −2: (−2, −1), (−2, 0), (−2, 3) — vertical

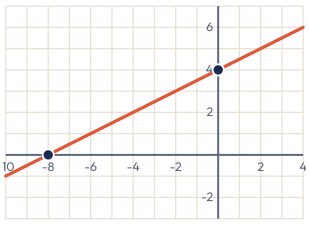
One horizontal, one vertical line

TYPE 3 Graph by using intercepts

1. x-intercept: replace y with 0 and solve. y = 0.
2. y-intercept: replace x with 0 and solve. x = 0.
3. Plot both points and draw the line. ارسم.

SOLVED EXAMPLE

Graph −x + 2y = 8.



y = 0: −x = 8 → x = −8 — (−8, 0)

x = 0: 2y = 8 → y = 4 — (0, 4)

Line through (−8, 0) and (0, 4)

Draw the Line

Graphing Linear Functions

Book 4-1 · p.209–218 | 4 question types | Linear and Nonlinear Functions

SCAN FOR PRACTICE

MUST KNOW · CONTINUED

TYPE 4 Use intercepts in context

1. Find both intercepts.
احسب المقطعين.
2. y-intercept = the value at the start (**x = 0**).
المقطع الصادي = البداية.
3. x-intercept = when the quantity reaches 0.
المقطع السيني = لما يخلص.

SOLVED EXAMPLE

$y + \frac{5}{2}x = 60$: cups of dog food y after x days.

$y = 0 \rightarrow \frac{5}{2}x = 60 \rightarrow x = 24$ — x-int

$x = 0 \rightarrow y = 60$ — y-int

60 cups at the start; empty after 24 days

DON'T FALL FOR IT

× WRONG
~~x-intercept: let x = 0~~

✓ RIGHT
let y = 0

On the x-axis, y is 0.
على محور x، الـ y = 0.

× WRONG
~~x = −2 is a horizontal line~~

✓ RIGHT
vertical line

All points share the same x.
نفس x ← رأسي.

× WRONG
~~$y = -\frac{1}{4}x + 3$: use x = 1, 2, 3~~

✓ RIGHT
use x = −8, −4, 0, 4, 8

Multiples of 4 give whole-number y-values.
مضاعفات المقام أسهل.

Last Look

Graphing Linear Functions

THE WHOLE LESSON ON ONE PAGE

1 Make a table
Choose 3–5 x-values, find y, plot, connect.
$y = -2x - 3$: (0, −3), (1, −5)
جدول، نقط، خط.

2 Good x-values
With a fraction slope, choose multiples of the denominator.
$y = \frac{1}{4}x + 3$: $x = -8, -4, 0, 4, 8$
اختار مضاعفات المقام.

3 y = a and x = a
$y = a$ horizontal; $x = a$ vertical (not a function).
$y = 5$ — · $x = -2$ |
أفقي ورأسي.

Linear Graphs

4 Intercepts
x-int: let $y = 0$. y-int: let $x = 0$. Plot both, draw the line.
$-x + 2y = 8 \rightarrow (-8, 0), (0, 4)$
صفّر واحد واحسب التاني.

5 Interpret
y-int = starting amount; x-int = when the amount reaches 0.
$y + \frac{5}{2}x = 60 \rightarrow$ 24 days
البداية والنهاية.

KEY WORDS · الكلمات المهمة

linear function	دالة خطية
x-intercept	المقطع السيني
y-intercept	المقطع الصادي
domain	المجال
solution of an equation in two variables	حل المعادلة

NIGHT-BEFORE CHECK

Q1 · BOOK STYLE
What are the intercepts of $-x + 2y = 8$?
A x-int 4, y-int −8
B x-int 8, y-int 4
C x-int −8, y-int 4
D x-int −8, y-int −4

Q2 · BOOK STYLE
The graph of $y = 5$ is...
A a line with slope 5
B the point (5, 5)
C a vertical line through (5, 0)
D a horizontal line through (0, 5)

Q3 · BOOK STYLE
For $4y = -12x + 36$, what is the x-intercept?

Q4 · LIKE BOOK P.213
$y + \frac{5}{2}x = 60$ gives the cups of dog food left after x days. After how many days is the bag empty?

Answers: Q1 C · Q2 D · Q3 3 · Q4 24

Rise Over Run

Rate of Change and Slope

Book 4-2 · p.219–228 | 4 question types | Linear and Nonlinear Functions

SCAN FOR PRACTICE

MUST KNOW

TYPE 1 Rate of change from a table

1. Pick two rows.
   اختار صفين.
2. Divide the change in y by the change in x.
   التغير في y ÷ التغير في x.
3. Give the units: y-units per x-unit.
   الوحدة.

SOLVED EXAMPLE

2 cups → 12 pancakes, 4 cups → 24 pancakes.

$$\frac{24-12}{4-2}=\frac{12}{2}$$

Answer: 6 pancakes per cup

TYPE 2 Is the function linear?

1. Find the rate of change between each pair of rows.
   احسب المعدل بين كل صفين.
2. All equal → linear. Not equal → not linear.
   كلهم زي بعض = خطية.

SOLVED EXAMPLE

x: 11, 8, 5, 2, −1 · y: −5, −3, −1, 1, 3

every step: $\frac{2}{-3}$ constant

Linear, rate $-\frac{2}{3}$

TYPE 3 Slope from two points

1. Label (x_1, y_1) and (x_2, y_2).
   سمّي النقط.
2. $m=\frac{y_2-y_1}{x_2-x_1}$. Keep the same order on top and bottom.
   نفس الترتيب.
3. Simplify the fraction.
   بسّط.

SOLVED EXAMPLE

(−3, 4) and (1, 7)

$$m=\frac{7-4}{1-(-3)}$$

$$=\frac{3}{4}$$

Answer: $\frac{3}{4}$

TYPE 4 Kinds of slope

1. Same y-values → horizontal → slope 0.
   نفس y ← صفر.
2. Same x-values → vertical → undefined.
   نفس x ← غير معرف.
3. Otherwise rising → positive, falling → negative.
   طالع موجب، نازل سالب.

SOLVED EXAMPLE

(3, 1) and (3, 6)

$m=\frac{5}{0}$ cannot divide by 0

Answer: undefined

Rise Over Run

Rate of Change and Slope

Book 4-2 · p.219–228 | 4 question types | Linear and Nonlinear Functions

SCAN FOR PRACTICE

DON'T FALL FOR IT

× WRONG

~~$m = \frac{x_2 - x_1}{y_2 - y_1}$~~

✓ RIGHT

$m = \frac{y_2 - y_1}{x_2 - x_1}$

Rise (y) on top, run (x) at the bottom.

y فوق و x تحت.

× WRONG

$\frac{7-4}{-3-1}$

✓ RIGHT

$\frac{7-4}{1-(-3)}$

Subtract in the same order on top and bottom.

نفس الترتيب.

× WRONG

~~vertical line → slope 0~~

✓ RIGHT

undefined

The run is 0 and you cannot divide by 0.

رأسي = غير معرف.

Last Look

Rate of Change and Slope

THE WHOLE LESSON ON ONE PAGE

1 Rate of change

$\frac{\text{change in y}}{\text{change in x}}$. y depends on x.

pancakes: $\frac{24-12}{4-2} = 6$ per cup

التغير في y على التغير في x.

2 Compare rates

Positive → increase; negative → decrease.
Bigger size → faster change.

$\frac{1325-1550}{5} = -45$ per year

موجب زيادة، سالب نقص.

3 Linear or not

Linear only if the rate of change is the same for every pair.

$\frac{3}{7}$ and $\frac{2}{7}$ → not linear

معدل ثابت = خطية.

Slope

4 Slope formula

$m = \frac{y_2 - y_1}{x_2 - x_1}$

(−3, 4), (1, 7) → $m = \frac{3}{4}$

الفرق في y على الفرق في x.

5 Four kinds

↗ positive · ↘ negative · — zero · | undefined

same x → undefined

موجب، سالب، صفر، غير معرّف.

KEY WORDS · الكلمات المهمة

rate of change	معدل التغير
slope	الميل
rise	التغير الرأسي
run	التغير الأفقي
undefined slope	ميل غير معرف

NIGHT-BEFORE CHECK

Q1 · BOOK STYLE

Find the slope through (−3, 4) and (1, 7).

Q2 · LIKE BOOK P.220

A budget was AED 1550 in 2010 and AED 1325 in 2015. Find the rate of change per year.

Q3 · BOOK STYLE

The slope of the line through (2, 5) and (2, −1) is...

A 0 B −3
C 3 D undefined

Q4 · LIKE BOOK P.221

Which table shows a linear function?

A x: 22, 29, 36 · y: −4, −1, 1
B x: 1, 2, 4 · y: 1, 2, 3
C x: 11, 8, 5 · y: −5, −3, −1
D x: 0, 2, 4 · y: 1, 4, 9

Answers: Q1 3/4 · Q2 −45 · Q3 D · Q4 C

Start and Steer

Slope-Intercept Form

Book 4-3 · p.229–238 | 4 question types | Linear and Nonlinear Functions

SCAN FOR PRACTICE

MUST KNOW

TYPE 1 Write an equation in slope-intercept form

1. Find m (slope) and b (y-intercept). حدد m و b.
2. Write **y = mx + b**. اكتب.

SOLVED EXAMPLE

Slope $\frac{4}{7}$, y-intercept 5.

$y = (\frac{4}{7})x + 5$

Answer: $y = \frac{4}{7}x + 5$

TYPE 2 Rewrite in slope-intercept form

1. Move the x-term to the right side. انقل حد x.
2. Divide every term by the coefficient of y. اقسم كل الحدود.

SOLVED EXAMPLE

Write $-22x + 8y = 4$ in slope-intercept form.

$8y = 22x + 4$ — add 22x

$y = \frac{22x + 4}{8}$ — divide by 8

Answer: $y = 2.75x + 0.5$

TYPE 3 Write a linear equation for a situation

1. Starting amount (**x = 0**) → b. البداية = b.
2. Change per unit → m. المعدل = m.
3. Define x and y. عرّف x و y.

SOLVED EXAMPLE

4.61 million job openings in May, increasing 0.06 million per month.

b = 4.61, m = 0.06

Answer: $y = 0.06x + 4.61$

Start and Steer

Slope-Intercept Form

Book 4-3 · p.229–238 | 4 question types | Linear and Nonlinear Functions

SCAN FOR PRACTICE

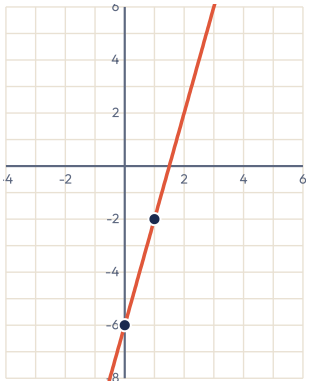
MUST KNOW · CONTINUED

TYPE 4 Graph from slope-intercept form

1. Rewrite in **y = mx + b** if needed.
   حوّل الأول.
2. Plot (0, b).
   ارسم (0, b).
3. Use rise over run to find another point; draw the line.
   استخدم الميل.

SOLVED EXAMPLE

Graph 12x − 3y = 18.



y = 4x − 6 — rewrite

(0, −6) → up 4, right 1 → (1, −2) — plot

Line through (0, −6) and (1, −2)

DON'T FALL FOR IT

× WRONG
~~8y = 22x + 4 → y = 22x + $\frac{4}{8}$~~

✓ RIGHT
y = 2.75x + 0.5

Divide every term by 8.
اقسم كل الحدود.

× WRONG
~~y = 4x − 6 → start at (0, 6)~~

✓ RIGHT
(0, −6)

b keeps its sign.
b بإشارتها.

× WRONG
~~m = $-\frac{3}{2}$: down 3 and left 2~~

✓ RIGHT
down 3, right 2

Only one of rise or run is negative.
سالب واحد بس.

Last Look

Slope-Intercept Form

THE WHOLE LESSON ON ONE PAGE

1 Write it
Put the slope in for m and the y-intercept for b.
$m = \frac{4}{7}, b = 5 \rightarrow y = \frac{4}{7}x + 5$
عوّض m و b.

2 Rewrite it
Solve for y: move the x-term, divide every term.
$-22x + 8y = 4 \rightarrow y = 2.75x + 0.5$
اعزل y.

$$y = mx + b$$

3 Real world
b = starting value; m = rate per unit.
$y = 0.06x + 4.61$
البداية والمعدل.

4 Graph it
Plot (0, b), then use rise over run for the next point.
$y = 4x - 6$: (0, −6) → (1, −2)
ابدأ من b واستخدم الميل.

5 Constant function
y = b: slope 0, horizontal.
$y = 3$
دالة ثابتة.

KEY WORDS · الكلمات المهمة

slope-intercept form	صيغة الميل والمقطع
parameter	وسيط / بارامتر
constant function	دالة ثابتة

NIGHT-BEFORE CHECK

Q1 · BOOK STYLE
Which is the equation of the line with slope −5 and y-intercept 12?
A $y = -5x - 12$ B $y = 12x - 5$
C $y = -5x + 12$ D $y = 5x + 12$

Q2 · BOOK STYLE
What is the slope-intercept form of $-16x - 4y = -56$?
A $y = -4x + 14$ B $y = 4x - 14$
C $y = -4x - 14$ D $y = 16x + 56$

Q3 · BOOK STYLE
Write $-22x + 8y = 4$ in slope-intercept form.
A $y = 22x + 4$ B $y = 2.75x + 0.5$
C $y = -2.75x + 0.5$ D $y = 2.75x + 4$

Q4 · LIKE BOOK P.230
183 million users in 2012, increasing by 9 million per year. Which equation fits (x = years after 2012)?
A $y = 183x + 9$ B $y = 9x - 183$
C $y = 192x$ D $y = 9x + 183$

Answers: Q1 C · Q2 A · Q3 B · Q4 D

Slide, Stretch, Flip

Transformations of Linear Functions

Book 4-4 · p.239–250 | 4 question types | Linear and Nonlinear Functions

SCAN FOR PRACTICE

MUST KNOW

TYPE 1 Translations

1. Number added outside → up / down.
برّه ← فوق/تحت.
2. Number inside with x → left / right (opposite sign).
جوه ← يمين/شمال بالعكس.

SOLVED EXAMPLE

Describe g(x) = (x − 6) + 3.

h = 6 → 6 right — inside

k = 3 → 3 up — outside

Answer: 6 units right and 3 units up

TYPE 2 Translations in context

1. Find what is added or subtracted after the main rule.
إيه اللي اتزود بعد الحساب.
2. Added → up; subtracted → down.
زيادة فوق، نقص تحت.

SOLVED EXAMPLE

f(t) = 12t; with a AED 4 fee g(t) = 12t + 4.

g(t) = f(t) + 4 — k = 4

Answer: 4 units up

TYPE 3 Dilations

1. a outside: |a| > 1 vertical stretch, 0 < |a| < 1 vertical compression.
برّه.
2. a inside: |a| > 1 horizontal compression, 0 < |a| < 1 horizontal stretch.
جوه بالعكس.
3. Stretch vertically / compress horizontally → steeper.
أشد انحدارًا.

SOLVED EXAMPLE

Describe $g(x) = (\frac{1}{4}x)$.

$g(x) = f(\frac{1}{4}x)$ — inside

$0 < \frac{1}{4} < 1$

Answer: horizontal stretch **(less steep)**

TYPE 4 Reflections

1. Negative outside: **−f(x)** → across the x-axis.
برّه ← حول x.
2. Negative inside: **f(−x)** → across the y-axis.
جوه ← حول y.

SOLVED EXAMPLE

Describe g(x) = −3x.

−1 · 3 · f(x)

Answer: reflected across the x-axis, stretched by 3

Slide, Stretch, Flip

Transformations of Linear Functions

Book 4-4 · p.239–250 | 4 question types | Linear and Nonlinear Functions

SCAN FOR PRACTICE

DON'T FALL FOR IT

× WRONG
~~g(x) = (x + 5) → 5 right~~

✓ RIGHT
5 left

Inside the bracket, the shift is opposite to the sign.
جوه بعكس الإشارة.

× WRONG
~~f(4x) → horizontal stretch~~

✓ RIGHT
horizontal compression

|a| > 1 inside squeezes the graph.
جوه |a| > 1 ← انضغاط.

× WRONG
~~a translation changes the slope~~

✓ RIGHT
the slope stays the same

A slide does not tilt the line.
الانسحاب مش بيغير الميل.

Last Look

Transformations of Linear Functions

THE WHOLE LESSON ON ONE PAGE

1 Vertical translation
g(x) = f(x) + k: up if **k > 0**, down if **k < 0**.
x − 2 → 2 down
k برّه: فوق/تحت.

2 Horizontal translation
g(x) = f(x − h): right if **h > 0**, left if **h < 0**.
(x + 5) → 5 left
جوه: عكس الإشارة.

Transformations

3 Vertical dilation
a · f(x): |a| > 1 stretch (steeper); 0 < |a| < 1 compression.
2x → stretch
برّه: تمدد أو انضغاط رأسي.

4 Horizontal dilation
f(ax): |a| > 1 compression; 0 < |a| < 1 stretch.
$f(\frac{1}{4}x)$ → stretch
جوه: العكس.

5 Reflections
−f(x) across the x-axis; **f(−x)** across the y-axis.
−3x
سالب برّه x، جوه y.

KEY WORDS · الكلمات المهمة

parent function	الدالة الأم
identity function	الدالة المحايدة
translation	انسحاب
dilation	تمدد
reflection	انعكاس
family of graphs	عائلة منحنيات

NIGHT-BEFORE CHECK

Q1 · BOOK STYLE
Describe **g(x) = x − 2** compared with **f(x) = x**.
A 2 units right
B 2 units left
C 2 units up
D 2 units down

Q2 · BOOK STYLE
Describe **g(x) = (x + 5)** compared with **f(x) = x**.
A 5 units left
B 5 units down
C 5 units right
D 5 units up

Q3 · BOOK STYLE
Describe **g(x) = (x − 6) + 3**.
A 6 left, 3 up
B 3 right, 6 up
C 6 right, 3 down
D 6 right, 3 up

Q4 · LIKE BOOK P.244
The graph of **g(x) = 6x** is a ___ of the parent function and is ___.
A translation; parallel
B vertical stretch; steeper
C horizontal stretch; less steep
D vertical compression; less steep

Answers: Q1 D · Q2 A · Q3 D · Q4 B

Same Step, Every Step

Arithmetic Sequences

Book 4-5 · p.251–258 | 5 question types | Linear and Nonlinear Functions

SCAN FOR PRACTICE

MUST KNOW

TYPE 1 Identify an arithmetic sequence

1. Find the difference between each pair of terms. احسب الفروق.
2. Same every time → arithmetic; give d. ثابت ← حسابية.

SOLVED EXAMPLE

Is 17, 14, 10, 7, 3 arithmetic?

−3, −4, −3, −4 differences

Answer: No (no common difference)

TYPE 2 Find the next terms

1. Find d. احسب d.
2. Add d to the last term, again and again. زوّد d.

SOLVED EXAMPLE

11, 7, 3, −1, ...

d = −4

−1 − 4 = −5, −5 − 4 = −9, −9 − 4 = −13

Answer: −5, −9, −13

TYPE 3 Write the nth term

1. Find a_1 and d. حدد a_1 و d.
2. Substitute into $a_n = a_1 + (n - 1)d$. عوّض.
3. Distribute and simplify. بسّط.

SOLVED EXAMPLE

−4, −1, 2, 5, ...

$a_n = -4 + (n - 1)(3)$ $a_1 = -4, d = 3$

$= -4 + 3n - 3$

Answer: $a_n = 3n - 7$

TYPE 4 Find a specific term

1. Write the nth-term formula. القانون.
2. Substitute the term number for n. عوّض n.

SOLVED EXAMPLE

Find the 16th term of −4, −1, 2, 5, ...

$a_n = 3n - 7$

$a_{16} = 3(16) - 7 = 48 - 7$

Answer: 41

Same Step, Every Step

Arithmetic Sequences

Book 4-5 · p.251–258 | 5 question types | Linear and Nonlinear Functions

SCAN FOR PRACTICE

MUST KNOW · CONTINUED

TYPE 5 Arithmetic sequences as linear functions

1. Start value = term 0 (or use a_1).
   قيمة البداية.
2. d = amount added each step = slope.
   d = الميل.
3. **f(n) = dn + start**; domain: whole numbers (dots).
   المجال أعداد كلية.

SOLVED EXAMPLE

Account: 525, 580, 635, 690 at months 0–3.

d = 55 — common difference

f(n) = 580 + (n − 1)55 = 55n + 525

Answer: **f(n) = 55n + 525**, **domain {0, 1, 2, ...}**

DON'T FALL FOR IT

× WRONG	× WRONG	× WRONG
~~a_n = 4 + 3n for 4, 1, 2~~	~~11, 7, 3 → d = 4~~	~~a_8 means a = 8~~
✓ RIGHT	✓ RIGHT	✓ RIGHT
$a_n = 3n - 7$	d = −4	the 8th term
Use **(n − 1)**, not n. مش (n − 1) n.	Next minus previous: **7 − 11 = −4**. التالي ناقص السابق.	The subscript is a position. الرقم الصغير = الترتيب.

Last Look

Arithmetic Sequences

THE WHOLE LESSON ON ONE PAGE

1 Is it arithmetic?
Subtract each term from the next. All the same → common difference d.
17, 14, 10, 7 → −3, −4, −3 ×
الفرق ثابت.

2 Next terms
Add d again and again.
11, 7, 3, −1 → −5, −9, −13
زوّد d.

3 nth term
$a_n = a_1 + (n - 1)d$, then simplify.
−4, −1, 2 → $a_n = 3n - 7$
القانون وبسّط.

Arithmetic Sequences

4 Find a term
Substitute n into the formula.
$a_{16} = 3(16) - 7 = 41$
عوّض n.

5 As a linear function
f(n) = dn + start; d is the slope; the graph is dots, not a line.
f(n) = 55n + 525
d = الميل، والرسم نقط.

KEY WORDS · الكلمات المهمة

sequence	متتابعة
term	حد
arithmetic sequence	متتابعة حسابية
common difference	الفرق المشترك
nth term	الحد النوني

NIGHT-BEFORE CHECK

Q1 · BOOK STYLE
Which is an arithmetic sequence?
A 2, 4, 8, 16, ...
B 1, 4, 9, 16, ...
C 82, 73, 64, 55, ...
D 17, 14, 10, 7, ...

Q2 · BOOK STYLE
The next three terms of 31, 18, 5, ... are...
A −8, −18, −31
B −8, −21, −34
C −13, −26, −39
D 8, −5, −18

Q3 · BOOK STYLE
Which equation gives the nth term of −4, −1, 2, 5, ...?
A $a_n = -4n + 3$
B $a_n = 3n - 4$
C $a_n = 3n - 7$
D $a_n = 3n + 7$

Q4 · BOOK STYLE
Find the 16th term of −4, −1, 2, 5, ...

Answers: Q1 C · Q2 B · Q3 C · Q4 41

Piece by Piece

Piecewise and Step Functions

Book 4-6 · p.259–266 | 4 question types | Linear and Nonlinear Functions

SCAN FOR PRACTICE

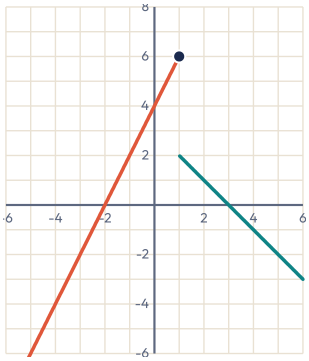
MUST KNOW

TYPE 1 Graph and evaluate a piecewise function

1. Find which interval the x-value belongs to.
حدد الفترة.
2. Use only that rule.
استخدم قاعدتها.
3. To graph: table for each piece; dot for ≤ / ≥, circle for < / >.
ارسم كل جزء بنقطته.

SOLVED EXAMPLE

f(x) = 2x + 4 if x ≤ 1; −x + 3 if x > 1. Graph and state the domain and range.



First piece ends at (1, 6) with a dot — 1 ≤ 1

Second piece starts at (1, 2) with a circle — x > 1

Domain: all real numbers · Range: y ≤ 6

TYPE 2 Greatest integer function

1. Work out the inside first.
احسب جوه.
2. Take the greatest integer ≤ that value (round down, also for negatives).
انزل، حتى مع السالب.

SOLVED EXAMPLE

Graph f(x) = ⟦x + 1⟧.

x = −0.25 → ⟦0.75⟧ = 0

x = 1.5 → ⟦2.5⟧ = 2

Steps of ⟦x⟧ moved 1 unit left

TYPE 3 Step functions in real life

1. Find the size of each step.
حجم الدرجة.
2. Any part of a step counts as a whole step (round up).
الجزء = درجة كاملة.
3. Make a table of intervals and values.
جدول فترات.

SOLVED EXAMPLE

1 lifeguard per 60 swimmers. How many for 130 swimmers?

0 < x ≤ 60 → 1 · 60 < x ≤ 120 → 2 · 120 < x ≤ 180 → 3 — table

Answer: 3 lifeguards

Piece by Piece

Piecewise and Step Functions

Book 4-6 · p.259–266 | 4 question types | Linear and Nonlinear Functions

SCAN FOR PRACTICE

MUST KNOW · CONTINUED

TYPE 4 Domain and range

1. Domain: all the x-values covered by the pieces.
المجال: كل قيم x.

2. Range: look at the lowest and highest y of every piece; dots are included, circles are not.
المدى: من الرسم.

SOLVED EXAMPLE

f(x) = 2x + 4 (x ≤ 1), −x + 3 (x > 1)

every x is used → all real numbers — domain

highest point (1, 6) dot; both pieces go down forever — range

Range: y ≤ 6

DON'T FALL FOR IT

× WRONG
~~⟦2.75⟧ = 3~~

✓ RIGHT
2

Always round down.
دايمًا لتحت.

× WRONG
~~⟦−0.25⟧ = 0~~

✓ RIGHT
−1

−1 is the greatest integer that is ≤ −0.25.
مع السالب: −1.

× WRONG
~~x > 1 → solid dot at x = 1~~

✓ RIGHT
open circle

> does not include 1.
< دايرة فاضية.

Last Look

Piecewise and Step Functions

THE WHOLE LESSON ON ONE PAGE

1 Piecewise function
Different rules on different x-intervals. Graph each piece on its own interval.
2x + 4 if x ≤ 1; −x + 3 if x > 1
كل قاعدة على فترتها.

2 Dots and circles
≤ or ≥ → solid dot (included). < or > → open circle.
(1, 6) dot · (1, 2) circle
≥ نقطة مصمتة، > دايرة فاضية.

Piecewise & Step

3 Evaluate
Check which interval x is in, then use that rule.
f(1) = 2(1) + 4 = 6
اختار الجزء الصح.

4 Greatest integer
⟦x⟧ = greatest integer ≤ x. Round down, never to the nearest.
⟦2.75⟧ = 2, ⟦−0.25⟧ = −1
انزل لأقرب صحيح أصغر.

5 Step function in real life
Any part of a unit counts as a whole unit → round up.
61 swimmers → 2 lifeguards
الجزء = وحدة كاملة.

KEY WORDS · الكلمات المهمة

piecewise-defined function	دالة متعددة التعريف
piecewise-linear function	دالة خطية متعددة التعريف
step function	دالة درجية
greatest integer function	دالة أكبر عدد صحيح

NIGHT-BEFORE CHECK

Q1 · BOOK STYLE
f(x) = 2x + 4 if **x ≤ 1**; **−x + 3** if **x > 1**. Find **f(1)**.

Q2 · BOOK STYLE
The range of **f(x) = 2x + 4 (x ≤ 1)**, **−x + 3 (x > 1)** is...

A **y ≥ 6**
B **y ≤ 6**
C **y ≤ 2**
D all real numbers

Q3 · BOOK STYLE
Evaluate **⟦−1.75⟧**.

Q4 · LIKE BOOK P.262
1 lifeguard is required for every 60 swimmers. How many for 245 swimmers?

Answers: Q1 · 6 · Q2 B · Q3 −2 · Q4 5

The V Shape

Absolute Value Functions

Book 4-7 · p.267–280 | 4 question types | Linear and Nonlinear Functions

SCAN FOR PRACTICE

MUST KNOW

TYPE 1 Translations of absolute value functions

1. k (outside the bars) → up if +, down if −.
   برّه.
2. h (inside) → |x − h| moves right, |x + h| moves left.
   جوه بالعكس.
3. Vertex = (h, k).
   الرأس.

SOLVED EXAMPLE

Describe g(x) = |x − 2| + 3.

h = 2 → 2 right — inside

k = 3 → 3 up — outside

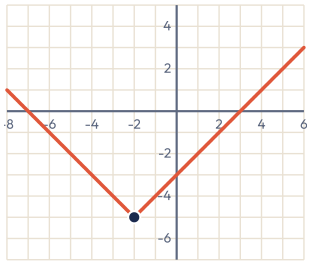
Answer: 2 right, 3 up, vertex (2, 3)

TYPE 2 Write the equation from a graph

1. Find the vertex (h, k).
   حدد الرأس.
2. Opens up → a positive; opens down → negative.
   الفتحة.
3. Write a|x − h| + k.
   اكتب.

SOLVED EXAMPLE

Vertex 2 units left and 5 units down, opens up.



g(x) = |x − (−2)| + (−5)

Answer: g(x) = |x + 2| − 5

TYPE 3 Dilations and reflections

1. a|x|: |a| > 1 vertical stretch (narrower); |a| < 1 compression (wider).
   برّه.
2. |ax|: |a| > 1 horizontal compression; |a| < 1 horizontal stretch.
   جوه.
3. A negative outside reflects across the x-axis.
   السالب ← مقلوبة.

SOLVED EXAMPLE

Describe $q(x) = -\frac{3}{4}|x|$.

negative → reflected across the x-axis

$\frac{3}{4} < 1$ → vertical compression

Answer: reflected across x, compressed vertically

The V Shape

Absolute Value Functions

Book 4-7 · p.267–280 | 4 question types | Linear and Nonlinear Functions

SCAN FOR PRACTICE

MUST KNOW · CONTINUED

TYPE 4 Evaluate an absolute value function

1. Work inside the bars first.
   جوه الأول.
2. Take the absolute value (make it non-negative).
   القيمة المطلقة.
3. Multiply by a, then add k.
   اضرب واجمع.

SOLVED EXAMPLE

$j(x) = -|x + 3| + 5$, find $j(-5)$.

$|-5 + 3| = |-2| = 2$

$-2 + 5$

Answer: 3

DON'T FALL FOR IT

× WRONG
~~$|x + 7|$ → 7 right~~

✓ RIGHT
7 left

$|x + 7| = |x - (-7)|$, so h = −7.
h = −7.

× WRONG
~~$|2x|$ → vertical stretch~~

✓ RIGHT
horizontal compression

a is inside the bars.
جوه = أفقي.

× WRONG
~~$g(x) = |x + 1| - 1$ is the same as $|x|$~~

✓ RIGHT
it moves 1 left and 1 down

The two shifts do not cancel.
الانسحابين مش بيلغوا بعض.

Last Look

Absolute Value Functions

THE WHOLE LESSON ON ONE PAGE

1 Parent function
$f(x) = |x|$: V shape, vertex (0, 0), domain all reals, range $y \geq 0$.
$|-3| = 3$
شكل V والرأس عند الأصل.

2 Translations
k outside → up/down; h inside → right if $x - h$, left if $x + h$. Vertex (h, k).
$|x - 2| + 3 \rightarrow$ vertex $(2, 3)$
الرأس (h, k).

$$a|x - h| + k$$

3 Equation from a graph
Read the vertex, check if it opens up or down.
vertex $(-2, -5) \rightarrow |x + 2| - 5$
اقرأ الرأس.

4 Dilations
$a|x|$: vertical. $|ax|$: horizontal (opposite effect).
$\frac{5}{2}|x|$ stretch · $|2x|$ horizontal compression
بزّه رأسي، جوه أفقي.

5 Reflections
$-a|x|$ opens down (across the x-axis).
$-|x + 3| + 5$: down, vertex $(-3, 5)$
السالب برّه = مقلوبة.

KEY WORDS · الكلمات المهمة

absolute value function	دالة القيمة المطلقة
vertex	الرأس
parent function	الدالة الأم
reflection across the x-axis	انعكاس حول x

NIGHT-BEFORE CHECK

Q1 · BOOK STYLE
Describe $g(x) = |x| - 3$ compared with $f(x) = |x|$.
A 3 units left
B 3 units up
C 3 units right
D 3 units down

Q2 · BOOK STYLE
The vertex of $g(x) = |x - 2| + 3$ is...
A (2, −3)
B (−2, 3)
C (3, 2)
D (2, 3)

Q3 · BOOK STYLE
Which describes $p(x) = |2x|$?
A translation 2 right
B horizontal compression
C vertical compression
D horizontal stretch

Q4 · LIKE BOOK P.269
A graph is the parent graph moved 2 units left and 5 units down. Its equation is...
A $g(x) = |x - 2| - 5$
B $g(x) = |x - 5| + 2$
C $g(x) = |x + 2| + 5$
D $g(x) = |x + 2| - 5$

Answers: Q1 D · Q2 D · Q3 B · Q4 D